Predicting Food Recall Costs using Environmental, Geographical, and Temporal Factors

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Target Audience

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Executive Summary

Public health agencies face challenges in anticipating spikes in pathogen-related costs, limiting their ability to allocate resources efficiently. We developed a two-stage machine learning framework using environmental data and FDA recall information to model both the likelihood and magnitude of pathogen-related events across U.S. states.

In the first stage, an XGBoost classification model predicts the probability of recall events, achieving strong performance (ROC AUC ≈ 0.85–0.99 depending on feature set). In the second stage, a regression model estimates the severity of costs conditional on an event. While event occurrence is highly predictable, severity shows minimal predictability ($R^2 \approx 0.03$), indicating that cost variability is driven by factors not captured in the dataset.

We recommend using the classification model as an early-warning decision support tool to guide proactive interventions, inspection prioritization, and resource allocation. The results highlight that environmental conditions contribute to event risk, but operational and structural factors dominate cost outcomes.

Problem

Public health costs tied to foodborne pathogens are hard to predict. The data are sparse and highly skewed, and many of the biggest drivers, like supply chain issues or operational failures, aren't directly captured. Because of this, models that try to predict costs directly tend to struggle, and agencies often end up reacting after outbreaks happen instead of getting ahead of them.

Objective & Approach

To address this, we break the problem into two parts: first, predicting when a recall event is likely to happen, and second, estimating how costly that event might be. This matches how the data actually behaves, events are more structured and predictable, while costs are much noisier. In practice, we find that events can be predicted with high accuracy using patterns over time and across states, but cost remains difficult to estimate in all cases. This means the model is most useful as an early warning tool, helping identify when and where risk is elevated so agencies can prioritize inspections and allocate resources more effectively.

Background

Food recall risk is shaped by many factors, including production practices, supply chain conditions, regulatory processes, and potentially environmental conditions. In this project, we focus on temperature and precipitation because prior research suggests they may influence pathogen presence, transport, and growth, making them reasonable candidate predictors for food safety risk.

Rainfall has been proposed as a mechanism for pathogen transport through runoff, particularly following long dry periods when contaminants accumulate on land and are then flushed into water systems (MDPI, 2025). Studies have linked heavy precipitation to elevated levels of pathogens such as *Salmonella* and *E. coli* in agricultural and coastal environments (Mavian et al., 2020).

Temperature is commonly associated with pathogen dynamics, influencing growth rates, survival, and ecological conditions. For example, higher sea surface temperatures have been correlated with cholera outbreaks through increases in plankton populations that host *Vibrio cholerae* (Constantin de Magny et al., 2008). Recent work also suggests that warmer temperatures may accelerate horizontal gene transfer, potentially contributing to antimicrobial resistance (Smith & Roberts, 2025).

Dataset & Exploratory Data Analysis

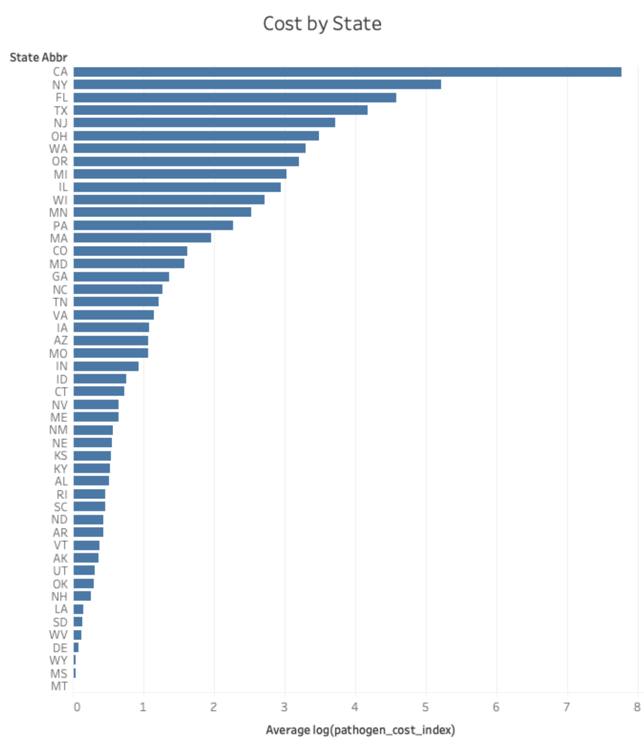
We constructed a panel dataset combining environmental data with public health and regulatory records to study food safety risk over time. Weather data were sourced from the NOAA National Climatic Data Center (nClimDiv), providing monthly state-level measures of temperature and precipitation. These were merged with FDA Food Enforcement Reports (via openFDA), which track all food product recalls in the U.S.

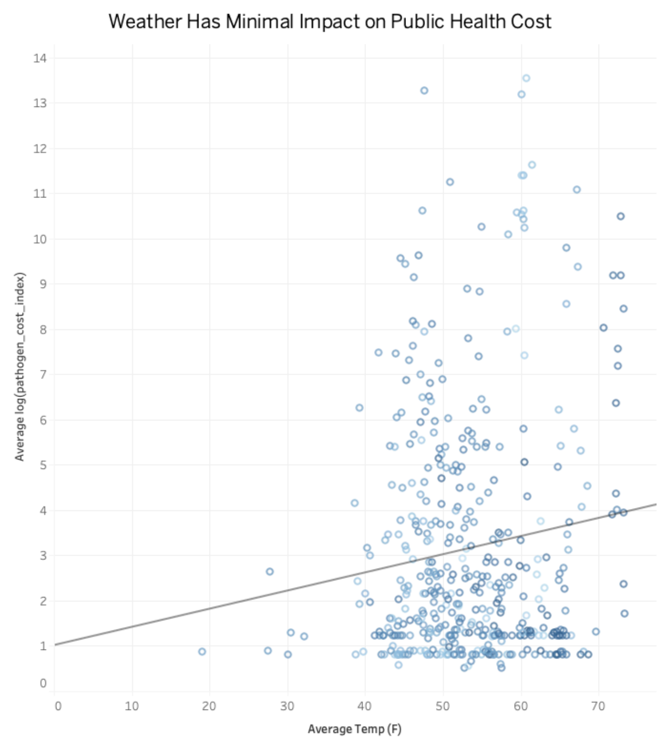
Using regex-based text extraction, we identified pathogen types in recall descriptions and linked them to cost estimates from the USDA Economic Research Service (ERS). These estimates are based on CDC incidence data and convert illness counts into economic impact using two approaches:

cost-of-illness (COI), which captures medical and productivity losses, and willingness-to-pay (WTP), which incorporates the Value of Statistical Life (VSL) for fatal outcomes.

To model recall risk, we engineered features across four main groups:

- **State baseline features** capture persistent differences across states, including average cost levels and historical event rates.
- **Event history features** capture short-term dynamics, such as whether a recall occurred in the previous month and how frequently events have occurred recently.
- **Seasonality features** represent time-of-year effects using month and cyclical transformations (sine and cosine) to capture repeating annual patterns.
- **Weather features** include temperature and precipitation levels, along with lagged values, rolling averages, deviations from normal conditions, and interaction terms to capture short-term trends and combined environmental effects.





The visualizations highlight a clear contrast between geographic and environmental influences on public health cost. Costs vary substantially across states, with states such as California, New York, and Florida exhibiting significantly higher average costs, indicating that location is an important driver of variation. In contrast, the relationship between temperature and cost is weak, with a low R^2 and widely dispersed data points, even when accounting for precipitation. Together, these results suggest that geographic and structural factors play a more meaningful role in explaining cost variation than environmental conditions.

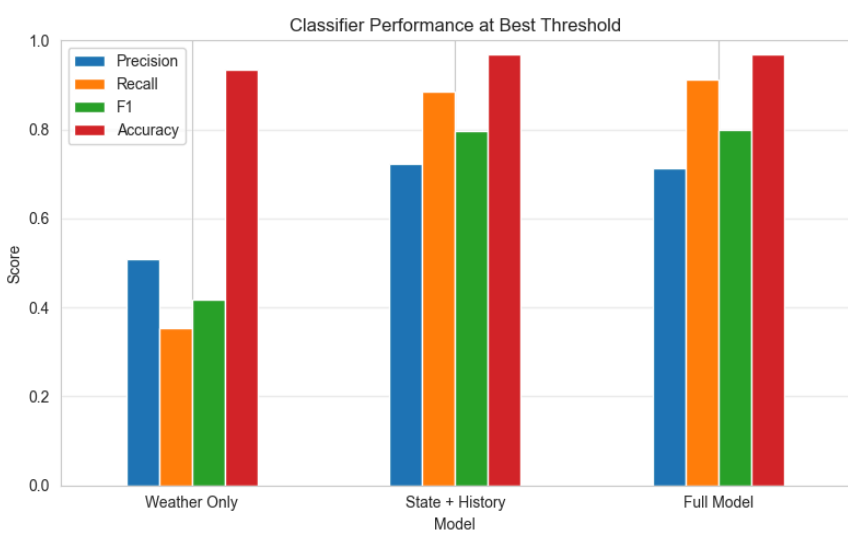
Methodology

We model pathogen-related risk using a two-stage approach that separates event occurrence from cost severity. In the first stage, we use XGBoost classification to predict the probability of a recall event at the state-month level. To understand what drives predictive performance, we evaluate three feature sets: (1) weather-only features, (2) state, temporal, and event-history features, and (3) a combined model including both. Models are trained using time-series cross-validation to preserve temporal ordering and evaluated on a held-out 2025 test set using ROC AUC and average precision.

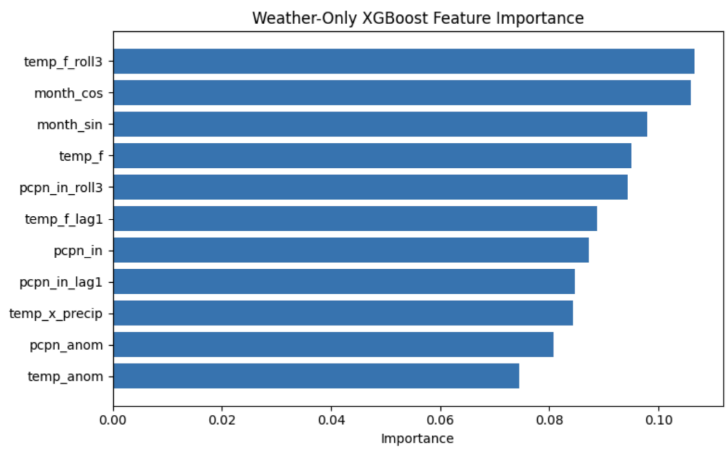
In the second stage, we model cost severity using XGBoost regression on event-only observations, estimating the magnitude of cost conditional on an event occurring. This avoids diluting the signal with zero-cost observations and better reflects the structure of the data. Performance is evaluated using RMSE, MAE, and R^2. This framework allows the classification model to capture structured patterns in when events occur, while the regression model tests whether those same features can explain variation in cost severity.

Results

The classification performance highlights a clear tradeoff between precision and recall across feature sets. The weather-only model performs substantially worse, with low recall (0.354) and F1 score (0.418), indicating it misses a large portion of true events. In contrast, the state and history model significantly improves both precision (0.722) and recall (0.886), resulting in a much higher F1 score (0.795). The full model provides only marginal gains, slightly increasing recall (0.911) at a small cost to precision (0.713), with nearly identical overall performance (F1 = 0.800). This suggests that most of the predictive signal comes from temporal and state-level patterns, with weather contributing only minor incremental value.

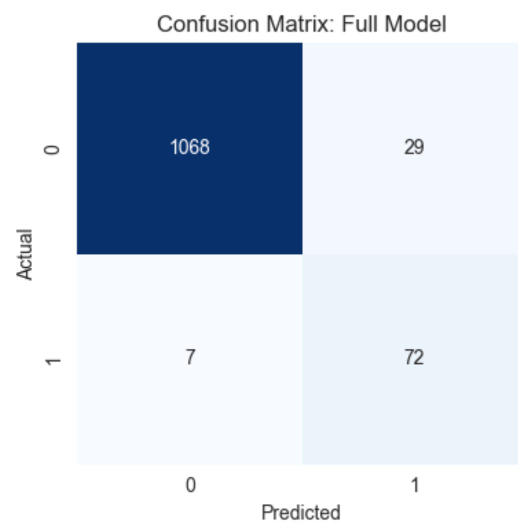


The confusion matrix for the full model reinforces this interpretation. The model correctly captures the majority of true events while maintaining strong performance on non-event observations, though it introduces some false positives to achieve high recall. Comparing results across models, both the state and history model and the full model achieve nearly identical AUC (0.989) and similar average precision (0.853 vs. 0.858), while far outperforming the weather-only model (AUC = 0.842, AP = 0.344). Overall, these results confirm that geographic and temporal features dominate predictive performance, while environmental variables add limited additional explanatory power.



The feature importance plot from the weather-only XGBoost model indicates that temperature and precipitation-related variables contribute relatively evenly to predictions, with no single dominant driver. Rolling temperature (temp_f_roll3), seasonal components (month_sin and month_cos), and precipitation measures (pcpn_in_roll3 and pcpn_in) all show comparable importance levels, suggesting the model is capturing general environmental patterns. However, despite this distribution of feature importance, overall model performance remains weak (R^2 = 0.033, ROC AUC = 0.842, MAE = 2.75), indicating that weather variables collectively explain very little of the variation in public health cost. This suggests that while environmental factors contain some structure, they have minimal practical predictive value when used in isolation.

The state and history model alone achieves near-perfect accuracy (ROC AUC ≈ 0.99), indicating that recall events are largely explained by persistent state-level risk and temporal clustering. The weather-only model performs substantially worse (ROC AUC ≈ 0.84), showing that environmental variables alone provide limited signal.



Adding weather features to the full model does not improve overall discrimination relative to the state and history baseline. However, it does slightly shift the precision–recall tradeoff, improving precision at the cost of recall. This suggests that weather information may help the model be more selective (reducing false positives), but does not meaningfully increase its ability to identify additional true events.

	Model	AUC	AP	Best Threshold	Precision	Recall	F1
0	Weather Only	0.842	0.344	0.680	0.509	0.354	0.418
1	State + History	0.989	0.853	0.868	0.722	0.886	0.795
2	Full Model	0.989	0.858	0.873	0.713	0.911	0.800

In contrast, the severity models show very limited predictive power. Even with the full feature set, the regression model explains only a small fraction of cost variability ($R^2 \approx 0.03$ on event-only observations). Predictions tend to regress toward the mean and fail to capture large cost spikes. This gap suggests that event occurrence follows stable, learnable patterns, while cost severity is driven by factors not captured in the dataset, such as contamination scale, distribution reach, and supply chain dynamics.

Conclusion

This analysis demonstrates that pathogen-related public health risk can be effectively decomposed into two components: occurrence and severity. Environmental conditions, geographic variation, and temporal patterns provide strong predictive signals for the likelihood of recall events, enabling the development of accurate early-warning models.

However, the magnitude of associated costs remains largely unpredictable using these features. This suggests that cost outcomes are driven primarily by operational and structural factors not captured in environmental or recall data.

From a policy perspective, these findings support the use of classification models for proactive risk monitoring and resource allocation. Public health agencies can leverage these models to identify high-risk periods and locations, even though precise cost forecasting remains infeasible with the available data.

Future work should focus on integrating supply chain, firm-level, and contamination-specific data to improve the modeling of cost severity.

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